Conditions for a linear model: A closer look at the residuals

MATH 213

To use linear regression, the following conditions should hold:

- **Linearity:** The relationship between explanatory and response should be linear.

- **Uniform spread of the residuals:** Residuals should exhibit constant variability. That is, the variability of points around the least squares line should be roughly constant.

- **No patterns in the residuals.**

- **Normality of residuals:** The distribution of residuals should be “nearly normal”, that is mostly mound-shaped and symmetric in shape.

Head length vs Total length for Brushtail Possum

Linearity: Check the scatterplot for evidence of a linear relationship.

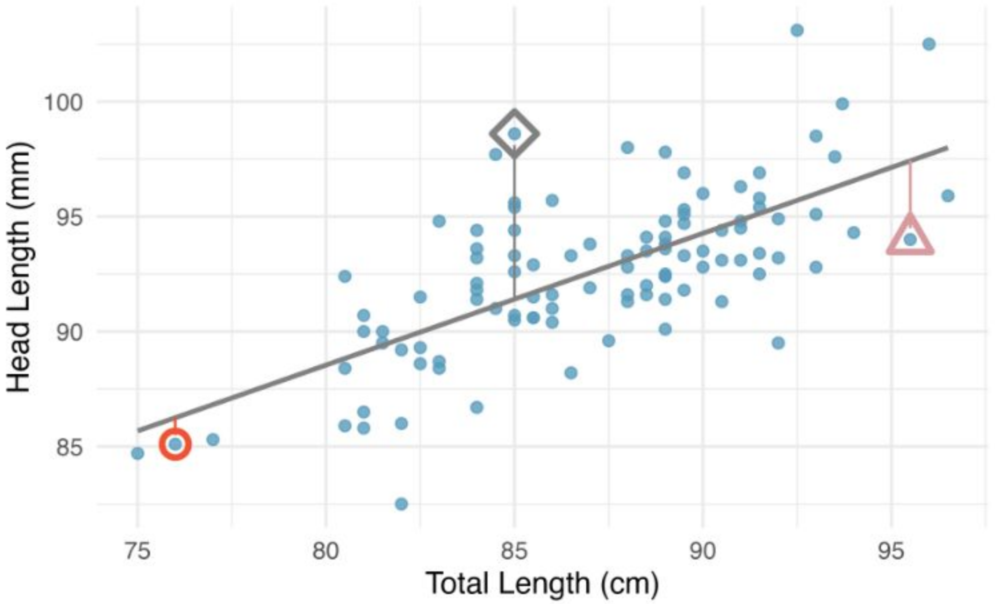

Figure 7.8: A reasonable linear model was fit to represent the relationship between head length and total length, with three points highlighted.

Residual plot

This is called a "residual plot". It is a plot of the residuals versus the predicted values of head length.

To use linear regression, we need to ensure that **residuals are evenly distributed about the horizontal line at residual = 0**, that the **variability is approximately constant**, and that there are **no patterns** in the residuals.

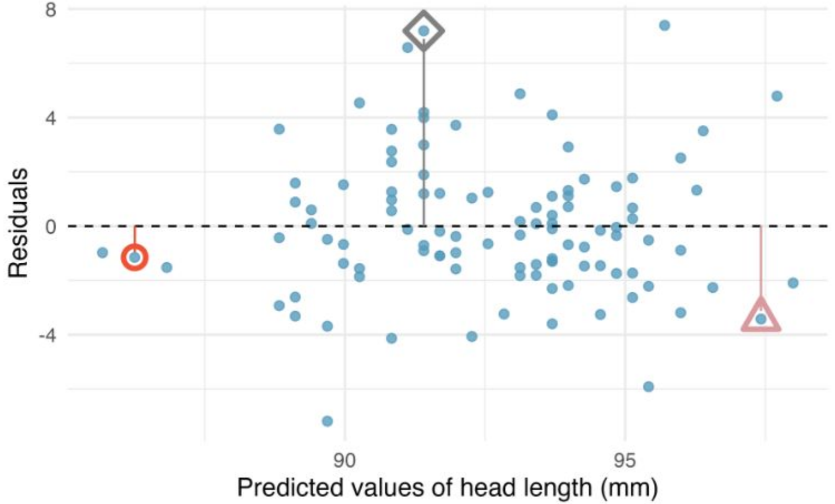

Figure 7.9: Residual plot for the model predicting head length from total length for brushtail possums.

https://openintro-ims.netlify.app/model-slr.html#resids

Residual plots, ctd.

One purpose of residual plots is to identify characteristics or patterns still apparent in data after fitting a model. Figure 7.10 shows three scatterplots with linear models in the first row and residual plots in the second row. Can you identify any patterns remaining in the residuals?

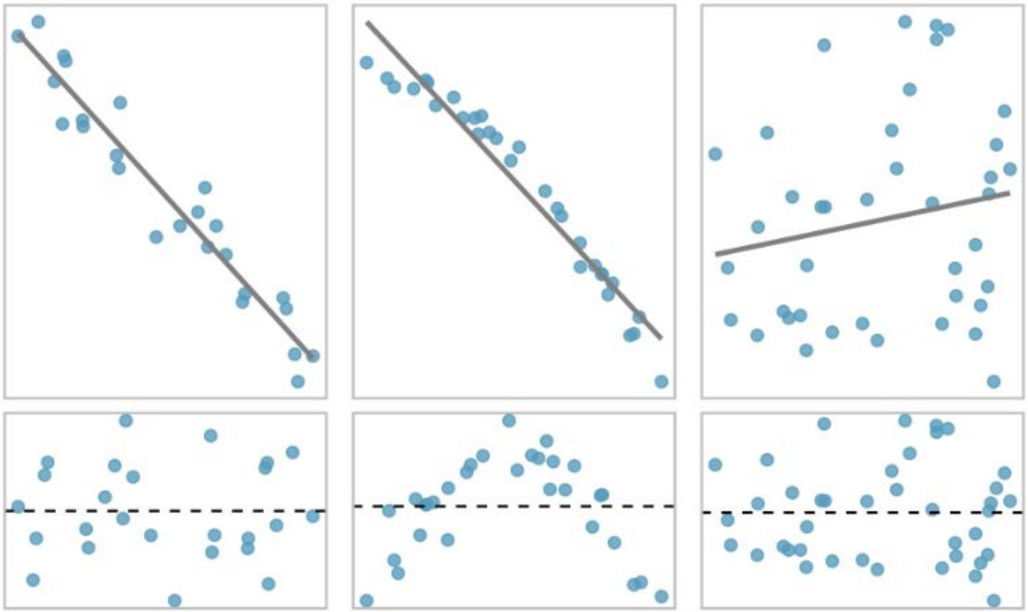

Figure 7.10: Sample data with their best fitting lines (top row) and their corresponding residual plots (bottom row).

Normality

To check this condition, make a histogram of the residuals and check to see that it is mound-shaped, symmetric, and centered around 0. Here is the histogram of residuals for our possums model.